1) a) $\binom{14}{4} = \frac{14 \cdot 13 \cdot 12 \cdot 11}{1 \cdot 2 \cdot 3 \cdot 4} = 77 \cdot 13 = 1001$ ✓

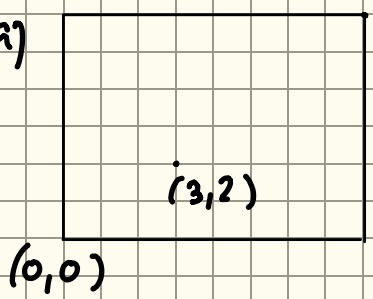
b) $\binom{6}{4} = \frac{6 \cdot 5 \cdot 4 \cdot 3}{1 \cdot 2 \cdot 3 \cdot 4} = 15$ $\qquad \binom{8}{4} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{1 \cdot 2 \cdot 3 \cdot 4} = 70$

$1001 - 85 = 916$ ✓

c) — — — — $\qquad \binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{1 \cdot 2 \cdot 3} = 56$ $\qquad \binom{8}{4} = 70$

$56 \cdot 6 = 336 \Rightarrow 336 + 70 = 106$ ✓

2) a)



$\binom{5}{3} \cdot \binom{9}{5} = \frac{5 \cdot 4 \cdot 3}{1 \cdot 2 \cdot 3} \cdot \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}$

$= 21 \cdot 60 = 1260$ ✓

b) $\binom{14}{6} - 1260 = \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = 77 \cdot 39 - 1260$

$= 1743$ ✓

c) $\binom{5}{2} \cdot \binom{4}{2} \cdot \binom{5}{2} = \frac{5 \cdot 4}{1 \cdot 2} \cdot \frac{5 \cdot 4}{1 \cdot 2} \cdot \frac{4 \cdot 3}{1 \cdot 2} = 10 \cdot 10 \cdot 6 = 600$ ✓

4) BANANA $\Rightarrow \binom{6}{1,3,2} = \frac{6!}{1! \cdot 3! \cdot 2!} = \frac{6 \cdot 5 \cdot 4}{2} = 60$

$\binom{5}{1,3,1} = \frac{5!}{3!} = 4 \cdot 5 = 20$

5) a) $\binom{9}{6} = \frac{9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3} = 84$

b) $\binom{10}{6} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = 210$

c) $\left(\!\binom{9}{6}\!\right) = \binom{14}{6} = \frac{14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = 3003$

6) $x+y+z=15$ a) $\left(\!\binom{3}{15}\!\right) = \binom{17}{15} = \frac{17 \cdot 16}{1 \cdot 2} = 136$

b) $y+z=15 \Rightarrow \left(\!\binom{2}{15}\!\right) = \binom{16}{13} = 16-2=15$

$15 \cdot 3 = 45 \Rightarrow 96-45=91$

c) $\left(\!\binom{3}{15}\!\right) \cdot \left(\!\binom{3}{14}\!\right) \cdot \ldots \cdot \left(\!\binom{3}{0}\!\right) \leq \binom{3}{1} =$

$\binom{17}{15} \cdot \binom{16}{14} \cdot \binom{15}{13} \cdot \ldots \cdot \binom{2}{0}$ 2

7) a) $\left(\!\binom{3}{10}\!\right) = \binom{12}{10} = \frac{11 \cdot 12}{2} = 66$

b) $\left(\!\binom{3}{4}\!\right) = \binom{6}{4} = \frac{6 \cdot 5}{2} = 15$

c) $\{1,2,\ldots,10\} \to \{1,2,3\}^4 = 3^{10}$

8) $\binom{8}{3,3,2} = \frac{8!}{3!3!2!} = \frac{4 \cdot 5 \cdot 6 \cdot 7 \cdot 8}{1 \cdot 2 \cdot 3 \cdot 2} = 10 \cdot 56 = 560$

$(x+y+z)^8$

b) $x+y+z=8$ $/-3$ $\left(\!\binom{3}{5}\!\right) = \binom{7}{5} = \frac{7 \cdot 6}{1 \cdot 2} = 21$

c) 3^8

Seminar

1) a) $\binom{49}{6} = \frac{49 \cdot 48 \cdot 47 \cdot 46 \cdot 45 \cdot 44}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = \dots$

b) Category 2: $\binom{6}{5} \cdot \binom{43}{1} = 6 \cdot 43 = 258$

Category 3: $\binom{6}{1} \cdot \binom{43}{2} =$

Category 4: $\binom{6}{3} \cdot \binom{43}{3} =$

Category 1: 1

c) $\binom{49}{6} - \underline{I} - \underline{II} - \underline{III} - \underline{IV}$

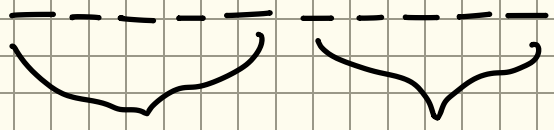
3) a)

6	4	$\binom{10}{6} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3 \cdot 4} = 210$
7	3	$\binom{10}{7} = \frac{10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3} = 120$
8	2	$\binom{10}{8} = \frac{10 \cdot 9}{2} = 45$
9	1	$\binom{10}{9} = 10$
10	0	1

$\Rightarrow 330 + 56 = 386$

b) — — — $\binom{5}{3} \cdot 2^5 = 10 \cdot 32 = 320$

c)



$\binom{5}{0}^2+\binom{5}{1}^2+\binom{5}{2}^2+\binom{5}{3}^2+\binom{5}{4}^2+\binom{5}{5}^2$

$= 1+25+100+100+25+1 = 252$

5) a $\binom{8}{2,3,1,1,1,2,1,1}$ MATHEMATICIAN

MATHEICN → 1

$\frac{8!}{3!\cdot 2!\cdot 2!} = \frac{4\cdot 5\cdot 6\cdot 7\cdot 8}{4}$ $= 30\cdot 56 = 1680 : 2 = 840$

b) $A,A,A,E,I,I \Rightarrow \binom{12}{6}\cdot\binom{7}{2,2,1,1,1} =$

c) A,A,A,E,I,I
C,H,M,M,N,T,T $\binom{12}{6} =$

4) 12 : 3F, 9B

9) $\binom{9}{3}\cdot\binom{6}{3}\cdot\binom{3}{3} = 1680$

$\overline{F}$ $\overline{F}$ $\overline{F}$

b) $\frac{1}{2}\binom{12}{6} = \frac{12!}{6!\cdot 6!}\cdot\frac{1}{2} = 462$

c) $\left(\!\binom{12}{6}\!\right) = \binom{16}{5} = 4368$

6) a) $a+b+c+d = 7$, $a \neq 0$

$a' = a-1 \geq 0 \Rightarrow a'+1 = a \Rightarrow a'+b+c+d = 6$

$\Rightarrow \left(\left(\begin{matrix}4\\6\end{matrix}\right)\right) = 84$

b) $a+b+c+d=11$
$a'+b+c+d=10$??

7) $(x-2y+2)^6$ a) $x+y+z=6 \Rightarrow \left(\left(\begin{matrix}3\\6\end{matrix}\right)\right) = \binom{8}{6}$
$= \frac{8\cdot 7}{1\cdot 2} = 28$

b) $\binom{6}{2,3,1} = \frac{6!}{3!2!1!} = \frac{4\cdot 5\cdot 6}{2} = 60$

$(-2y)^3 = -8y^3 = 0-480$

c) $(1-2\cdot 1+1)^6 = 0$ d) $|-2y| = 2y$
$\Rightarrow (1+2+1)^6 = 4^6$

8) $\underline{1}_\underline{1}_\underline{1}_\underline{1}_\underline{1}_\underline{1}_$ $(0,0,0)$ (0)

a) $\binom{10}{2} = \frac{10\cdot 9}{2} = 45$ $(0,0)$ $(0,0)$

$\binom{10}{4} - \binom{10}{2} - \binom{10}{2} = \frac{10\cdot 9\cdot 8\cdot 7}{1\cdot 2\cdot 3\cdot 4} - 90$

$= 210 - 90 = 120$?!

b)

9) ???

10) a) K pers. $n-k \Rightarrow \left(\left(\begin{matrix}k\\n-k\end{matrix}\right)\right) = \binom{n-1}{n-k}$

b) $\binom{n}{k} = \binom{n+k-1}{n}$ c)

Model test:

1) $\underline{9}\ \underline{10}\ \underline{10}\ \underline{10} = 9000$

0, 1, 2, 3, 4, 5, 6, ~~7~~, 8, 9

$\underline{8}\ \underline{9}\ \underline{9}\ \underline{9} = 72 \cdot 81$

$81 \cdot 72$

162

567

5832

$\Rightarrow 3168$ (R: D) ✓

2) 0, 1, 2, 3, 4, 5, 6

$\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ } \Rightarrow 300$ (R: A) ✓

$\underline{1}\ \underline{1}\ \underline{1}\ \underline{1}$

$\downarrow\ \downarrow\ \downarrow\ \downarrow$

5 5 4 3

3) 4B, 46

1, 2, 3, 4 - 6

A, B, C, D - B

① ← A

③ 0 → 1

0 → 2

③ 0 → 3

① 0

$\downarrow$

4

$6 \cdot 4! = 144$ (R: A)

4) $\underline{\ }\ \underline{\ }\ \underline{\ }\ \underline{\ }$

6 6 5B

$\binom{11}{4} - \binom{6}{4}$

$= \frac{11 \cdot 10 \cdot 9 \cdot 8}{1 \cdot 2 \cdot 3 \cdot 4} - \frac{6 \cdot 5}{1 \cdot 2}$

$= 330 - 15 = 315$ (R: B) ✓

5) BALADA

C: B, L, D

V: A, A, A

B — $\binom{5}{3,1,1} = \frac{5!}{3!} \cdot 3 = 5 \cdot 4 = 20 \cdot 3$

D

L

$= \frac{5!}{2!} = 3 \cdot 4 \cdot 5 = 60$

R: E

6) A, B, C, D, E, F

8 — — — — — — — —

$\binom{6}{3} = \frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3} = 20$

3, 3, 2

2, 3, 3

3, 2, 3

A, B, C

$\binom{3}{8} = \binom{10}{8}$

$= \frac{10 \cdot 9}{2}$

$A + B + C = 8$

$45 \cdot 20 = 900$ R: E

A = 1

B = 1

C = 1

$\binom{3}{5} = \binom{7}{5} = \frac{7 \cdot 6}{2} = 21$

$21 \cdot 20 = 420$